Experiment No-19 : Student-Teacher-Subject Code

Aim:

To write a Prolog program to implement the STUDENT-TEACHER-SUB-CODE database and answer queries about students, teachers and subject codes.

Programming Code:

```
% ---------- Facts ----------
% student(RollNo, Name)
student(101, ravi).
student(102, priya).
student(103, arun).
student(104, divya).

% teacher(Name)
teacher(kumar).
teacher(meena).
teacher(suresh).

% subject(SubCode, SubjectName)
subject(cs101, 'Artificial Intelligence').
subject(cs102, 'Data Structures').
subject(cs103, 'Database Systems').
subject(cs104, 'Operating Systems').

% handles(Teacher, SubCode)
handles(kumar,  cs101).
handles(meena,  cs102).
handles(meena,  cs103).
handles(suresh, cs104).

% studies(RollNo, SubCode)
studies(101, cs101).
studies(101, cs102).
studies(102, cs101).
studies(102, cs103).
studies(103, cs104).
studies(104, cs102).
studies(104, cs104).

% ---------- Rules ----------
% Subject name taught by a teacher
subject_taught_by(T, SubName) :- handles(T, Code), subject(Code, SubName).

% Name of the subject for a given code
subject_name(Code, SubName) :- subject(Code, SubName).

% Code of the subject for a given name
subject_code(SubName, Code) :- subject(Code, SubName).

% Students who study a given subject code
students_in_subject(Code, Student) :- studies(Roll, Code), student(Roll, Student).

% Teachers who teach a given student
teacher_of_student(Student, Teacher) :-
    student(Roll, Student), studies(Roll, Code), handles(Teacher, Code).

% Students who share at least one subject
classmates(S1, S2) :-
```

```
    student(R1, S1), student(R2, S2), R1 \= R2,
    studies(R1, C), studies(R2, C).
```

OUTPUT:

```
$ swipl -q exp19.pl

?- subject_name(cs101, N).
N = 'Artificial Intelligence'.

?- subject_code('Data Structures', C).
C = cs102.

?- subject_taught_by(meena, S).
S = 'Data Structures' ;
S = 'Database Systems'.

?- students_in_subject(cs101, S).
S = ravi ;
S = priya.

?- teacher_of_student(ravi, T).
T = kumar ;
T = meena.

?- classmates(ravi, X).
X = priya ;
X = divya.

?- handles(suresh, cs101).
false.

$
```

Result

The STUDENT-TEACHER-SUB-CODE database was implemented successfully using Prolog. The program stores facts about students, teachers and subject codes, and the queries correctly return the subjects, teachers, students and subject codes.